Model interpretability

Data Science Week 2-DS-04

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Summary

the main terms you will see throughout this document:

PHQ-9	Patient Health Questionnaire , a 9-question form used to measure how severe a patient's depression is. Scored 0–27. Higher = more severe. (0–4 minimal, 5–9 mild, 10–14 moderate, 15–19 moderately severe, 20–27 severe.)
GAD-7	Generalised Anxiety Disorder scale — a 7-question form used to measure anxiety severity. Scored 0–21. Higher = more anxious. (0–4 minimal, 5–9 mild, 10–14 moderate, 15–21 severe.) Both PHQ-9 and GAD-7 are routinely used together in NHS assessments.
Epworth Score	A simple questionnaire that measures how sleepy a person feels during the day — for example, whether they doze off while reading or watching TV. Higher scores mean worse daytime sleepiness, which is often linked to poor sleep quality at night.
BMI	Body Mass Index — a number calculated from height and weight. Used as a rough indicator of whether someone's weight may be affecting their health. It has known limitations but is widely used as a starting point.
K-Means / Clustering	A computer method for sorting patients into groups based on how similar they are across multiple factors at once — not just one score, but depression, anxiety, age, sleep, and BMI all together. Think of it as automatic patient profiling.
Regression Model	A mathematical formula that looks at several pieces of information about a patient and uses them to estimate (predict) a specific outcome — in our case, their anxiety score. It learns the formula from patterns in historical data.

Structural Break	A point in time when a long-term trend suddenly changes direction or speed, rather than just drifting gradually. For example, worry about crime declining sharply after a specific year rather than slowly year by year.

Part-1

Analysis Predicting a Patient's Anxiety Level

1.Dataset

Dataset Use - 783 university students with depression and anxiety records

2. What question were we trying to answer

The information we have, about the patient includes the patients depression score the patients sleep quality the patients age and the patients gender. So I am trying to see if we can use the patients depression score the patients sleep quality the patients age and the patients gender to estimate how anxious the patient will be.

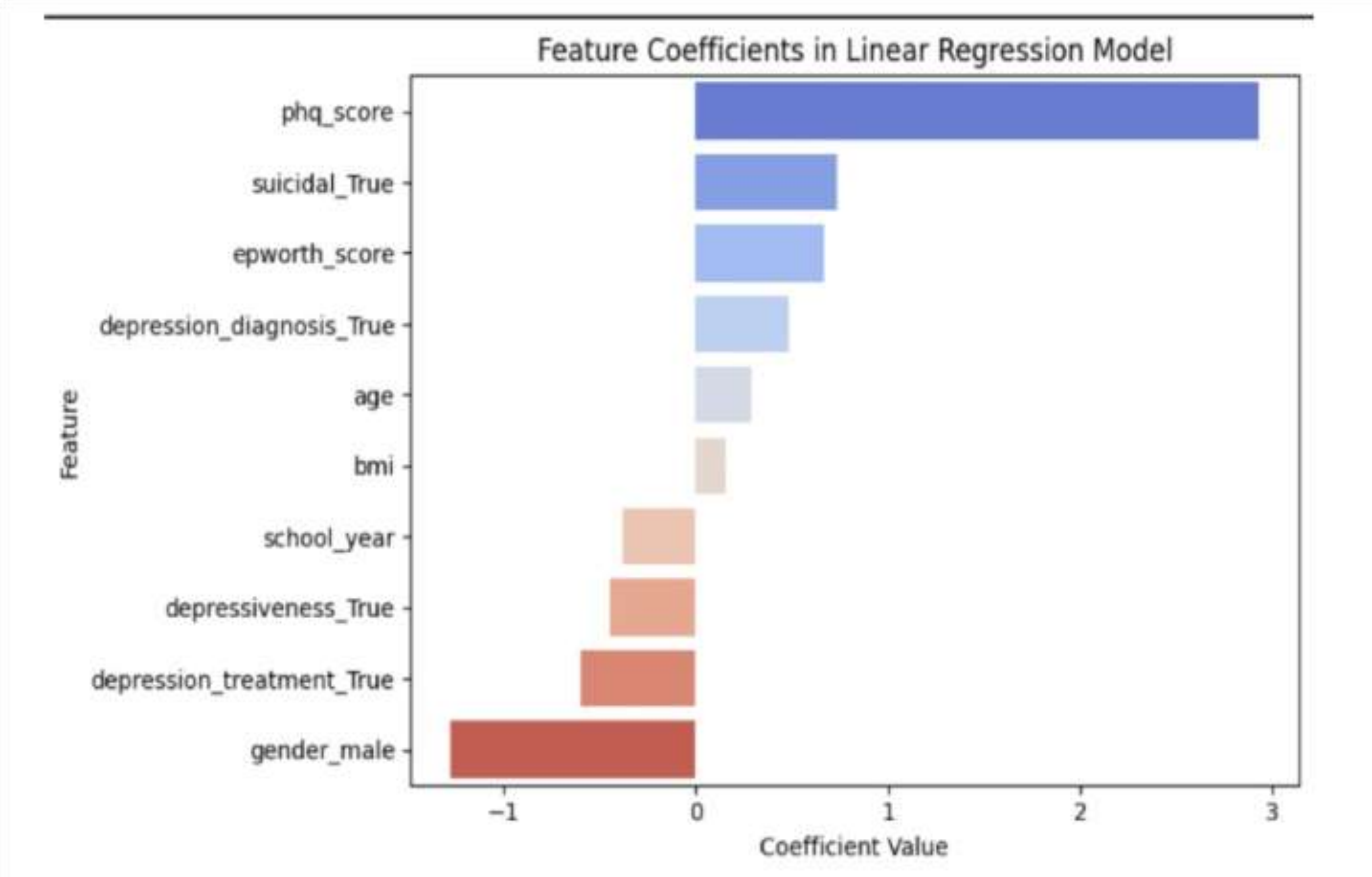
3. How does a 'prediction model' work.

Think of it like a recipe. The model is given hundreds of patient records where we know both the ingredients and the final dish. It learns which ingredients matter most and in what quantities. Once trained, it can take a new patient's ingredients and estimate their likely anxiety score.

4.What information was used

The model used the following patient information as inputs:

Patient Information Used	What It Measures
PHQ-9 score	How depressed the patient is
Epworth score	How sleepy/fatigued they feel during the day
Age	Patient's age in years
BMI	Body Mass Index - weight relative to height
Gender	Male or female
Suicidal ideation	Whether the patient has reported suicidal thoughts
Depression diagnosis	Whether they have a formal depression diagnosis
Depression treatment	Whether they are currently receiving treatment
Academic year	What year of study they are in



5. What did the model find

The chart below shows which patient factors push anxiety scores up or down, and by how much. This is the most important output of the regression model.

Each horizontal bar represents one patient factor. Bars pointing RIGHT (blue) mean that factor tends to increase anxiety. Bars pointing LEFT (red/orange) mean it tends to decrease anxiety.

Here is what each bar means in plain language:

Factor	Direction	Meaning
PHQ-9 (depression score)	Raises anxiety	By far the most powerful signal. A patient with a high depression score is very likely to also have high anxiety. The bar is more than 3 times longer than any other — this dominates everything else.

Suicidal ideation	Raises anxiety	Patients who have reported suicidal thoughts tend to score notably higher on anxiety. This is a strong clinical warning sign.
Epworth score (sleepiness)	Raises anxiety	Poor sleep and daytime fatigue push anxiety upward. Sleep is both a symptom of anxiety and something that makes anxiety worse — a vicious cycle.
Depression diagnosis	Raises anxiety	Having a formal depression diagnosis is linked to higher anxiety, though the effect is smaller than the raw PHQ score.
Age & BMI	Slightly raises	Small positive effects — older age and higher BMI are modestly associated with higher anxiety in this dataset.
Being male	Lowers anxiety	The strongest downward factor. Male patients scored considerably lower on anxiety than females. Important caveat: men may be less likely to report distress, not necessarily less likely to feel it.
Receiving depression treatment	Lowers anxiety	Patients already in treatment for depression tend to show slightly lower anxiety — a possible sign that care is helping.
Academic year & depressiveness flag	Slightly lowers	Small negative effects — more advanced students and those flagged as 'depressive' (but perhaps managing it) showed marginally lower anxiety.

6.Accuracy of the Model

The model correctly explained about 46% of the differences in anxiety scores between patients (this is measured by something called R^2, which you do not need to understand in detail — just know that 46% is meaningful for mental health data, where many things that affect anxiety, like home life and social support, were not in the dataset at all).

The avg error in model is about 3 points. Means if a patient score is 12 the model will give output between 9 to 15.

Limitations

- The data comes from university students only. Results may not apply to older NHS patients or non-student populations.
- The model cannot account for factors not in the dataset home situation, financial stress, relationship problems, all of which clinicians know matter enormously.
- An average error of 3 GAD-7 points means predictions are useful for flagging risk groups, but not precise enough to replace individual assessment.

Analysis 2 — Grouping Patients by Mental Health Profile

1.Dataset:

Dataset Use - 783 university students (775 after removing incomplete records)

2. Why we need this

Rather than predicting a single score, this analysis asked: are there natural 'types' of patient in our data groups of people who share similar patterns across depression, anxiety, age, BMI, and sleep all at once? If so, understanding those types could help tailor care more effectively.

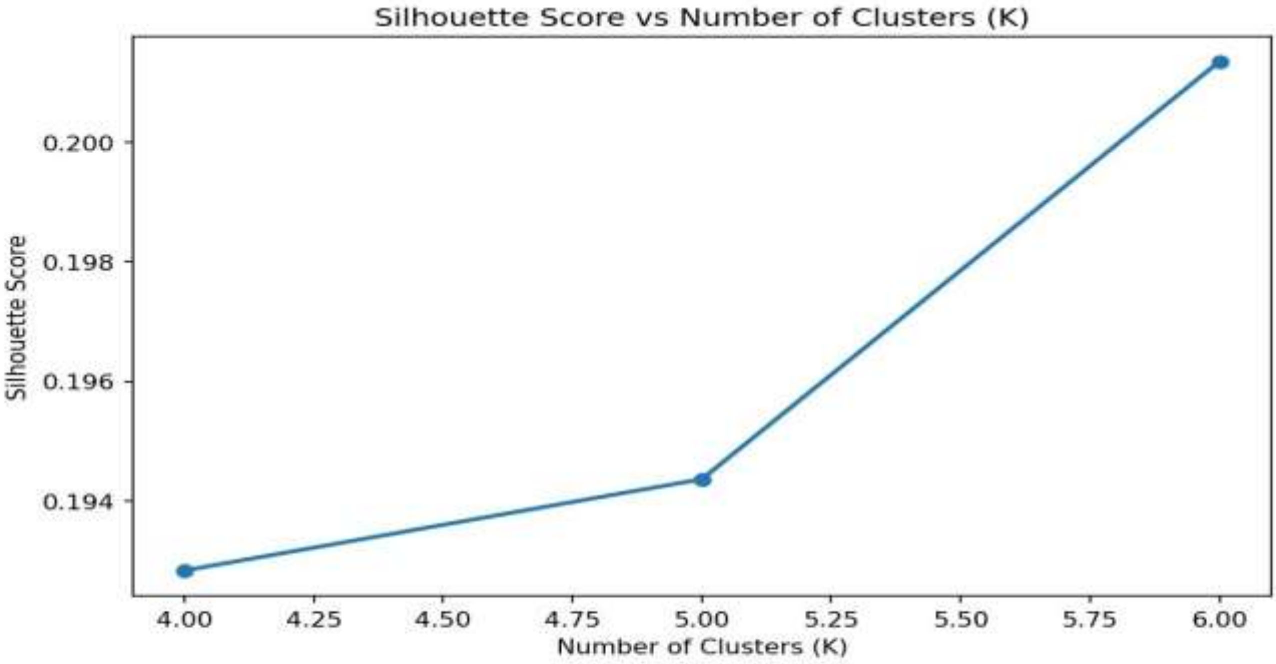
3.What is clustering and why does it matter clinically

Clustering is like sorting a pile of patient notes by hand but across six factors simultaneously rather than one. The computer finds groupings where patients within a group are very similar to each other, and quite different from patients in other groups.

For clinical teams, this means moving from 'this patient has a GAD score of 12' to 'this patient looks like our moderate-anxiety-with-sleep-problems group' which is more actionable because it suggests a type of care that has worked for similar patients.

4.How did we decide on 6 groups?

The team tested different numbers of groups (from 4 to 6) and used two checks to find the best number. The charts below show those checks:



5.The six patient groups

After sorting 775 patients into 6 groups, here is what each group looks like in plain language:

Group	Size	Profile Name	Key Characteristics
Group 0	259 patients (33%)	Low-Risk Males	Mostly male patients. Low depression, low anxiety, healthy sleep. This is the baseline low-risk population the 'doing well' group.
Group 1	56 patients (7%)	Obesity,Associated Mild Distress	Mild mental health symptoms but significantly elevated BMI (average ~31, which falls in the obese range). Suggests a connection between physical health and mental wellbeing worth exploring in care planning.
Group 2	167 patients (22%)	Moderate Anxiety & Depression with Sleep Problems	The largest at-risk group. Moderate depression and anxiety, but with notably worse sleep scores. Sleep disruption may be both a symptom and a driver here — a practical intervention target.
Group 3	61 patients (8%)	Older Students, Mild Distress	Slightly older students (average age ~24 vs ~19–20 in other groups) with mild symptoms. May reflect different pressures faced by mature or postgraduate students.
Group 4	46 patients (6%)	HIGH RISK ,Severe Anxiety & Depression	The most urgent group. Highest PHQ-9 scores, highest GAD-7 scores, highest daytime sleepiness. Only 46 patients but every single one of them warrants prioritised clinical attention.

Group 5	186 patients (24%)	Low-Risk Healthy Females	Mostly female patients with minimal distress, healthy BMI and good sleep. The female equivalent of Group 0.

a) Group 4 contains only 46 patients (6% of the sample) but has the most severe mental health profiles in the entire dataset. Average PHQ-9 of 17 (moderately severe depression) and GAD-7 of 16 (severe anxiety). These patients also have the worst sleep scores. Early identification and prioritised support for patients matching this profile could have the greatest clinical impact.

b) Clinical Takeaway *Six meaningful patient groups were identified. The largest at-risk group (Group 2, 167 patients) shows a clear sleep-anxiety connection sleep-focused support could be a practical, scalable entry point. Group 1's obesity link suggests integrated physicalmental health pathways are worth considering. Group 4 is small but clinically critical.*

6.Limitations

- Silhouette scores are low (around 0.20), meaning the groups overlap this is normal for mental health data and means the profiles are tendencies, not definitive categories.
- Group 4 has only 46 patients . due to small size we can’t give a sure ans about this group.
- This is a single snapshot in time. A patient's group could change as their condition evolves over weeks or months.
- The K-Means algorithm works best with round, equal-sized groups. Mental health data is rarely that tidy future work could explore more flexible grouping methods.
- The student-only dataset may not represent the full range of NHS patients.

Analysis 3 — How Public Worry About Crime Has Changed Over 30 Years

1.Dataset

Dataset Use - Office for National Statistics (ONS) Crime Survey, England and Wales, 1992–2025

2. Why we need this

What the uk public view over the safety of themselves over 30 years. And if so, were there specific years where the change happened suddenly rather than gradually. This would help for anxiety changes.

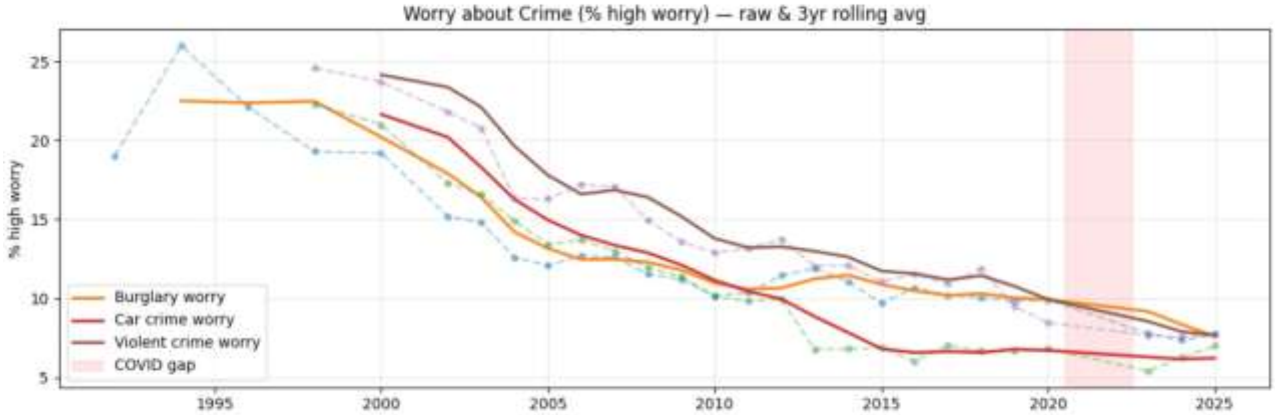
3. Dataset

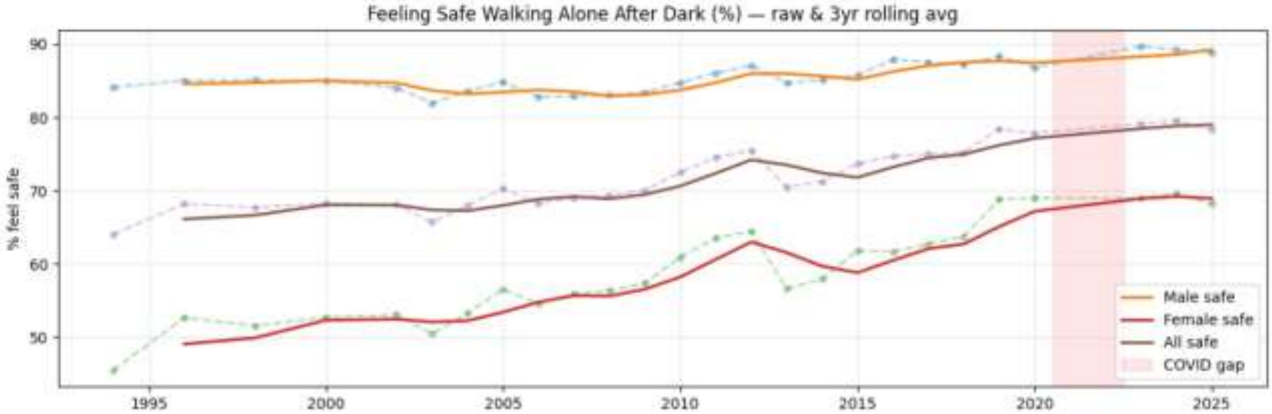
The ONS Crime Survey talks to thousands of people in England and Wales every year. They ask about their experiences and feelings about crime.

There are two types of information we looked at:

a) Worry about crime. This is the number of people who say they are very worried about things like burglary, car crime and violent crime.
b) Feeling safe at night. This is the number of people who feel walking alone in the dark.

The information goes back to 1992 still 2025There is a gap, in 2021 and 2022. This is because the survey was stopped during the COVID-19 pandemic





4 . Model Prediction

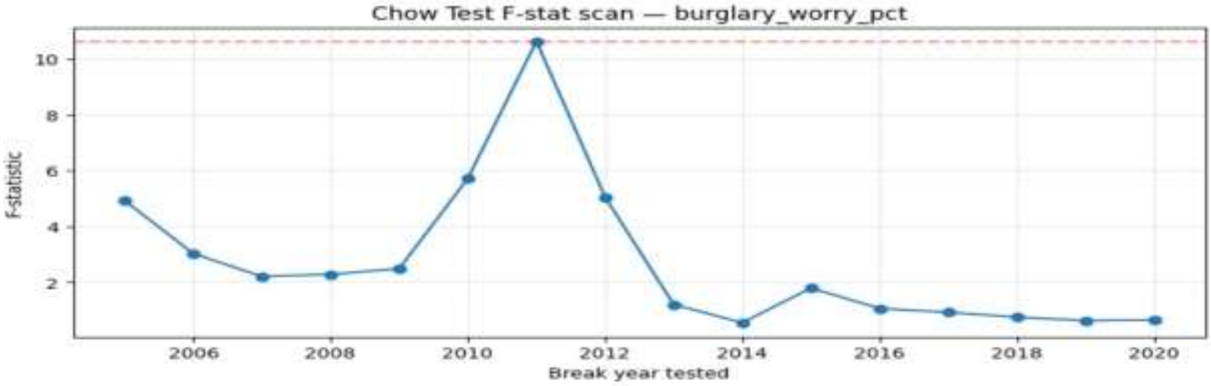
Three clear patterns were seen over the 30-year period:

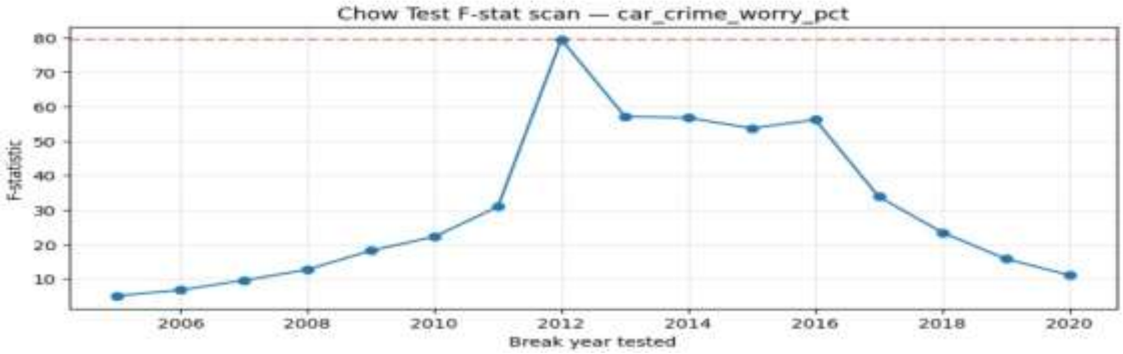
Trend	What Happened	What It Means
Crime worry falling	All three types of crime worry (burglary, car, violent) declined consistently from the mid-1990s through to the mid-2020s. Burglary worry dropped from around 26% to roughly 8%.	Public anxiety about crime, as a whole, has fallen dramatically over a generation. This is a genuine positive for community mental health.
People feel safer	The percentage of people feeling safe walking alone after dark increased over the same period, especially for women rising from around 45% to nearly 70% by the mid-2010s.	Safety perceptions have improved substantially, particularly for women. This represents a meaningful quality-of-life shift.
Gender gap narrowing	Women started from a much lower baseline of feeling safe, but showed the biggest improvement over time.	The gap between male and female safety perceptions has been
		narrowing, though women still feel considerably less safe than men.

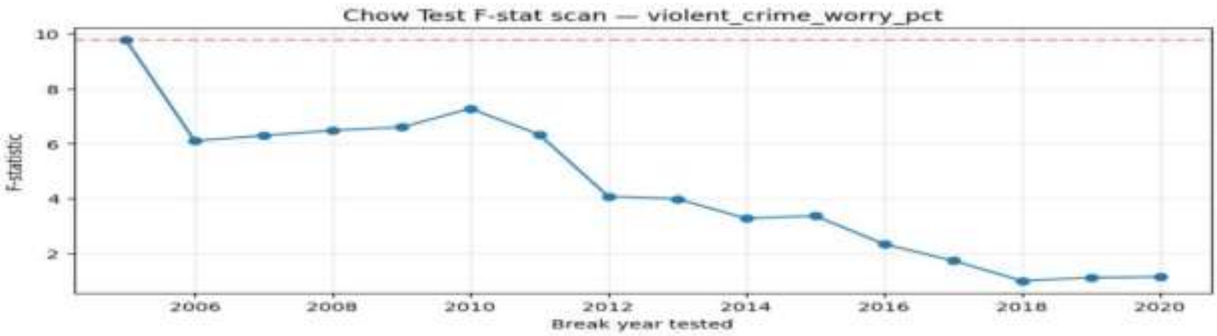
5. The Structural Break Analysis

We just looking at the overall trend, we used a special statistical test called a Chow test. This test helped them find years where the trend changed suddenly. The team used the Chow test to identify these changes in the trend.

Measure	Year of Sudden Change	How Strong?	Likely Explanation
Car crime worry	2012	Very strong	Worry dropped sharply, coinciding with a period of falling actual car crime rates and changes in how car crime was officially recorded.
Burglary worry	2011	Strong	A notable acceleration in the already-falling trend, coinciding with long-term declines in actual burglary rates.
Violent crime worry	2005	Strong	An earlier shift, possibly reflecting changing public awareness following media coverage of crime trends in the early 2000s.
Women feeling safe after dark	2012	Significant	A meaningful upturn in women's sense of personal safety — one of the most positive findings in the dataset.
Men / overall feeling safe	Around 2009–2012	Modest	An improving trend, but less sharp and abrupt than the change seen for women.







Result. This study does not tell doctors how to treat each patient. It helps them understand what is going on. If a patient is anxious because they are scared of crime or worried about their safety doctors can now use thirty years of information that shows crime and people worrying about crime have gone down a lot. For people who plan health services in communities they should look at areas where people are still very scared even though crime is not as high as it used to be and they might need to do something to help people, in those areas feel better.

6.Limitations

- This is national data for England and Wales. Local variation- in high-crime urban areas, for example - may look very different from the national average.
- People get worried about crime. This shows that a community is feeling anxious. This worry about crime is not the thing as being diagnosed with anxiety by a doctor.
- We had a break in the data because of COVID-19 from 2021 to 2022. This break makes it hard to know if the changes we saw during the pandemic are going to last or if they were a temporary thing.
- When people answer surveys about crime they say what they are comfortable saying loud. What they say can be influenced by what they see in the news and what people, around them think, not what they have experienced themselves with crime.

Final Summary

Across Week-1 we get this final result

Key Theme	What the Data Shows	Suggested Clinical Action
Depression and anxiety always travel together	The PHQ-9 score was the single strongest predictor of anxiety far more powerful than any other factor.	Always assess both depression (PHQ9) and anxiety (GAD-7) together. Treating one while ignoring the other is likely to be less effective.
Sleep is a consistent risk factor	Poor sleep appeared in the regression model as a key predictor, and also defined the largest at-risk patient group (Group 2, 167 patients).	Sleep-focused support such as sleep hygiene advice or talking therapies targeting insomnia ,may be a practical, scalable entry point for reducing anxiety.
Gender differences are real but need careful interpretation	Male patients reported lower anxiety. Women showed the biggest improvement in safety perceptions over time.	Lower reported anxiety in men should not be taken at face value men may be less likely to report distress. Ensure male patients are not under-assessed.
A small group carries the highest risk	Only 6% of the student sample (46 patients) fell into the high-risk cluster, but they had the most severe profiles across every measure.	Early identification of patients matching this high-risk profile , high depression, high anxiety, poor sleep is where targeted intervention can have the greatest impact.
Public fear has fallen dramatically	Three decades of national data show crime worry declining and safety perceptions improving, especially for women.	Patients whose anxiety is partly rooted in fear of crime or community safety can be reassured with evidence. Persistent fear despite low actual risk may indicate a specific clinical need.